

What Likelihood Means: From a Raw Product to a Bounded, Comparable Score

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Abstract

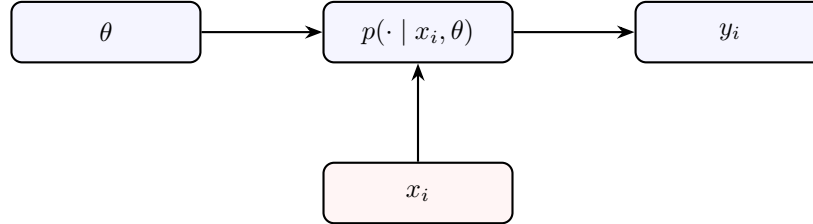
This note derives, from first principles, a single idea used across statistics, information theory and machine learning: read likelihood not as the raw product every textbook opens with, but as an exponentiated *average* log-density, $L := \exp(\mathbb{E}[\ln p])$. That reformulation is unusual: almost no standard reference defines likelihood this way. This note explains precisely why it earns its keep: an exact identity back to the textbook product, a direct information-theoretic reading (Jensen’s inequality, Shannon entropy, a temperature-indexed refinement borrowed from equilibrium statistical mechanics), two classical asymptotic results (Cramér’s large deviations and Wald’s martingale) that describe exactly this quantity’s long-run behavior, and a precise account of L as a statistical estimator, bias and variance included, together with a proof that the raw product it replaces is not an estimator of anything fixed at all. Two worked instantiations, a categorical classification model and a regression model under Gaussian noise, show the same construction landing on the reciprocal of a familiar error measure in both cases, perplexity in the first, root-mean-square error in the second. `elbow-helper`’s own mathematics note, `ELBOW-en.tex`, builds its Gaussian model-selection criteria (BIC, mBIC, ICL) on top of the foundation derived here rather than re-deriving it. Citations follow the same evidence-marking convention as `ELBOW-en.tex`: each claim states whether it is a verified citation, a citation reported through a secondary source, or this note’s own construction.

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1 The notion

Data here means n pairs (x_i, y_i) , an input x_i paired with an outcome y_i : a covariate driving a prediction, a position driving a curve’s height, or nothing at all, x_i simply absent, for a model that only ever describes y_i on its own. A single graphical model covers every case this note uses, θ and x_i feeding a conditional density that generates y_i :



Likelihood names the joint density of the observed outcomes, read as a function of the parameter rather than the data: $L_{\text{raw}}(\theta) := \prod_{i=1}^n p(y_i | x_i, \theta)$, a plain product of n terms, one per observation. This is the definition every standard treatment opens with, Bishop’s among the clearest (Bishop, 2006). It is also the exact quantity a Bayes factor or a Neyman-Pearson likelihood-ratio test compares directly. Left as a raw product, though, it is a poor building block for a bounded score: n densities multiplied together either shrink toward zero or explode as n itself grows, so L_{raw} on 40 points and L_{raw} on 4,000 points are not numbers a single threshold could ever compare; the raw product carries the sample size baked into its scale.

This note departs from that convention immediately and says so plainly rather than leaving the departure implicit inside a formula: it takes likelihood *per observation* from the outset, an exponentiated *average* log-density rather than a raw product,

$$L := \exp(\mathbb{E}[\ln p]) = L_{\text{raw}}^{1/n},$$

the geometric mean of the n individual densities, exactly the n -th root of L_{raw} , never a different number, only a different form of the same one. Almost no standard reference defines likelihood this way. The reason is not stylistic: dividing the exponent’s sum by n before re-exponentiating is exactly what turns an extensive quantity, one whose scale depends on sample size, into an intensive one, whose scale does not, the same move information theory makes when it works with *perplexity*, an exponentiated *average* log-probability, rather than a raw joint one (made literal in Section 4).

2 What the number means

Before any asymptotic machinery, L has a direct reading worth sitting with. By Jensen’s inequality, since $\ln$ is concave, $L = \exp(\mathbb{E}[\ln p]) \leq \mathbb{E}[p]$ always: the geometric mean this note works with never exceeds the arithmetic mean of the same densities, with equality only when p is constant across observations.

For a discrete p over K outcomes, take the expectation under p itself rather than under data: $\exp(\mathbb{E}_{y \sim p}[\ln p_y]) = \exp(-H(p))$, Shannon’s entropy (MacKay, 2003) read as a probability rather than a bit count. This sits in $[1/K, 1]$: equal to $1/K$ when p is uniform over the K outcomes, rising to 1 as p concentrates on one of them. Read this way, $\exp(-H(p))$ is a typical, or effective, probability. Its reciprocal $\exp(H(p))$ is the effective number of plausible outcomes: $\exp(H(p)) \approx 4$ means about as uncertain as a fair guess among four.

A temperature reading. That bound has a one-parameter refinement worth making precise rather than merely evocative. For $\beta > 0$, define the power-tilted family $p_\beta(y) := p(y)^\beta / Z(\beta)$,

$Z(\beta) := \int p(y)^\beta dy$ (or $\sum_y p(y)^\beta$ in the discrete case): a genuine exponential family, with β playing the role Gibbs assigns to inverse temperature against the “energy” $-\ln p(y)$. This is exactly the object the large-deviations statistical-mechanics dictionary describes: $\ln Z(\beta)$ is a scaled free energy, β its inverse temperature, and the Legendre-transform relationship Section 3’s Cramér argument uses for the rate function is, term for term, the one relating free energy and entropy in equilibrium statistical mechanics (Touchette, 2009). Differentiating at $\beta = 1$, where $p_\beta = p$, recovers L itself: $\frac{d}{d\beta} \ln Z(\beta)|_{\beta=1} = \mathbb{E}_p[\ln p] = \ln L$, so L is one derivative of a free energy, evaluated at one specific temperature. The Shannon entropy above is itself the $\beta \rightarrow 1$ limit of the Rényi entropy $H_\beta(p) := \ln Z(\beta)/(1 - \beta)$, a standard fact checked directly from the definitions above.

The two extremes of β read exactly like the two extremes of temperature, made precise rather than left as image. As $\beta \rightarrow 0^+$, hot and maximally agitated, p_β flattens toward uniform over p ’s support regardless of what p itself looked like: $Z(\beta) \rightarrow K$ in the K -category case, so $\exp(-H_\beta(p)) \rightarrow 1/K$, exactly the floor derived above. As $\beta \rightarrow \infty$, cold and frozen, p_β collapses onto p ’s own mode: $\exp(-H_\beta(p)) \rightarrow \max_k p_k$, the probability of the single most likely category, a tighter, p -dependent ceiling than the loose upper bound of 1 (reached only in the degenerate limit where p itself is already a point mass). Verified numerically on $p = (0.5, 0.3, 0.15, 0.05)$: $\exp(-H_\beta(p))$ moves from $0.25 = 1/4$ at $\beta \rightarrow 0$ to $0.5 = \max_k p_k$ at $\beta \rightarrow \infty$, monotonically, passing through $\exp(-H(p)) \approx 0.32$ at $\beta = 1$. Water’s own two familiar extremes at atmospheric pressure, freezing at 0°C and boiling at 100°C , are a physical illustration of exactly this shape, a fixed system swept from an ordered, low-entropy phase to a disordered, high-entropy one as an inverse-temperature-like parameter crosses its range, not a numerical correspondence: nothing here claims $\beta = 0$ or $\beta = \infty$ literally sit at those two Celsius values, only that the qualitative mechanism and its exact Legendre-transform bookkeeping are the ones equilibrium statistical mechanics already worked out.

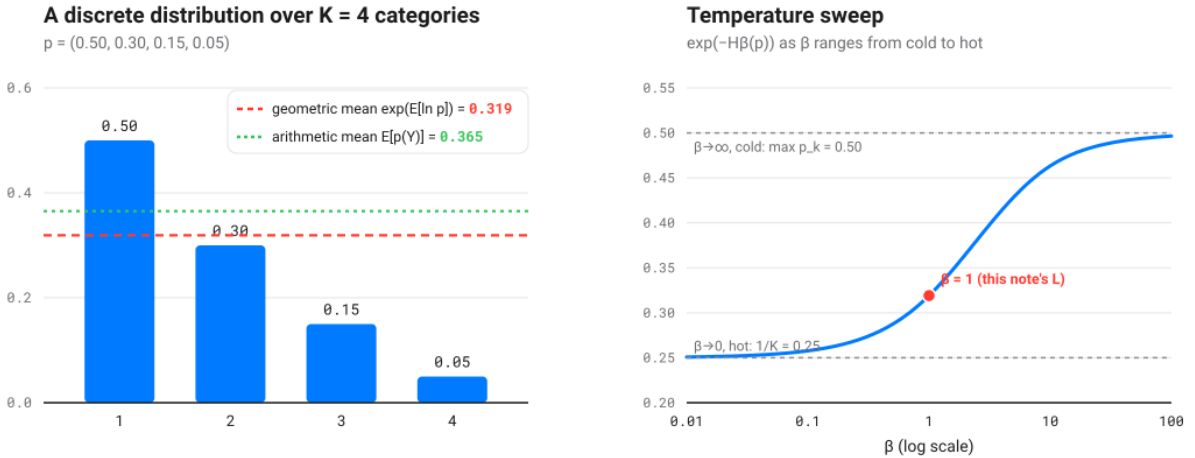


Figure 1: Left: the bar chart above, $p = (0.5, 0.3, 0.15, 0.05)$, with its geometric mean $\exp(\mathbb{E}[\ln p]) \approx 0.319$ and arithmetic mean $\mathbb{E}[p(Y)] \approx 0.365$ marked, Jensen’s inequality made visible as two horizontal lines rather than left as a bare formula. Right: $\exp(-H_\beta(p))$ swept across β , hot on the left, cold on the right, moving monotonically from the floor $1/K = 0.25$ to the ceiling $\max_k p_k = 0.5$, with $\beta = 1$, this note’s own L , marked where it sits along that path.

3 Why an average, not a sum: large deviations and martingales

Two classical results say this is not just a convenient rescaling. Write $Z_i := \ln p(y_i | x_i, \theta)$ for the per-observation log-density, so $L = \exp(\frac{1}{n} \sum_i Z_i)$, an exponentiated sample mean. First, the exact relation this buys back to L_{raw} : $L^n = \exp(\sum_i Z_i) = \prod_i p(y_i | x_i, \theta) = L_{\text{raw}}(\theta)$. Taking the n -th root and re-exponentiating an average does not lose information relative to the raw product; it is the same number, an n -th root away, now expressed in the one form whose large-sample behavior is understood.

Cramér’s theorem (Cramér, 1938), the founding result of large-deviations theory, says an i.i.d. sample mean like $\frac{1}{n} \sum_i Z_i$ does more than converge to $\mathbb{E}[Z]$, the law of large numbers’ guarantee; the probability that it strays from that limit by more than any fixed amount decays exponentially in n , at a rate set by a convex rate function, the Legendre transform of Z ’s cumulant generating function. Working with the average first, rather than the raw product L^n it is one root away from, is what exposes that exponential structure instead of burying it inside an already-multiplied number.

Chernoff (Chernoff, 1952) specializes exactly this machinery to deciding between two densities p_0, p_1 from n i.i.d. draws, in the symmetric setting where both hypotheses carry a prior and a single test minimizes the total (Bayes) probability of error rather than one error type at a fixed level of the other: the best total error probability any such test can achieve decays as $\exp(-n C(p_0, p_1))$, where the *Chernoff information* $C(p_0, p_1) := -\min_{\lambda \in [0,1]} \ln \int p_0^\lambda p_1^{1-\lambda}$ is, structurally, one more exponentiated average of a log, this time an integral over the sample space rather than a mean over observations. This is a different question from the one the Chernoff-Stein lemma answers, fixing one error type’s probability and asking how fast the other can be driven down, which lands on the Kullback-Leibler divergence rather than the Chernoff information; the two results are often confused precisely because both carry Chernoff’s name.

The same running product has a second, complementary life as a *process* rather than a single number. Define the likelihood-ratio process $\Lambda_t := \prod_{i=1}^t p_1(y_i)/p_0(y_i) = \exp(\sum_{i=1}^t \ln[p_1(y_i)/p_0(y_i)])$ over the first t observations. Under p_0 , Λ_t is a nonnegative martingale, $\mathbb{E}_{p_0}[\Lambda_t | \mathcal{F}_{t-1}] = \Lambda_{t-1}$, the fact Wald (Wald, 1945) built the Sequential Probability Ratio Test on: a threshold crossed by Λ_t at a data-dependent stopping time still controls the test’s error rate, by the optional stopping theorem, the guarantee a fixed- n test gets automatically and a naively-stopped one does not. That guarantee needs two fully specified densities p_0 and p_1 , though, a requirement not every sequential problem can meet; `elbow-helper`’s own sequential test (`ELBOW-en.tex`, the sequential-test section, `sec:fwer`) is exactly such a case, its alternative at each step a composite, best-fitting extra breakpoint rather than a second fixed density, which is why it calibrates against a permutation null instead of Wald’s classical martingale bound.

Two different questions about L as an estimator. Everything computed so far, L included, comes from a finite sample, so two distinct questions are worth separating cleanly rather than letting either one hide inside the other. First: is the sample-based L this note computes on n real observations a good estimator of the population value it converges to as $n \rightarrow \infty$? Second: does the raw, un-rooted product L_{raw} , this note’s own L before the n -th root is taken, estimate that same population value at all? Fix the target first. Let $(X_1, Y_1), \dots, (X_n, Y_n)$ be drawn i.i.d. from the true joint data distribution q . Let p_θ be a candidate model for $p(y | x)$. As $n \rightarrow \infty$, this note’s own L is built to converge to one fixed number,

$$L_\infty := \exp(\mu_Z), \quad \mu_Z := \mathbb{E}_q[\ln p_\theta(Y | X)],$$

not a random quantity: μ_Z is an ordinary expectation under the true q , a single number once p_θ and q are fixed. Two different sample-based quantities can now be compared against that one target.

The first is this note’s own L , computed on n observed pairs: $\hat{L}_n := \exp(\bar{Z}_n)$, $\bar{Z}_n := \frac{1}{n} \sum_i Z_i$, $Z_i := \ln p_\theta(Y_i | X_i)$. $\bar{Z}_n$ is exactly unbiased for μ_Z , so $\hat{L}_n$ is a genuine estimator of L_∞ , though not an unbiased one: $\exp$ is convex, so Jensen’s inequality, the same one Section 2 used to bound L itself, gives $\mathbb{E}[\hat{L}_n] \geq L_\infty$ always. The delta method makes the size of that upward bias explicit,

$$\mathbb{E}[\hat{L}_n] = L_\infty \left(1 + \frac{\sigma_Z^2}{2n} + O(n^{-2})\right), \quad \text{Var}[\hat{L}_n] = L_\infty^2 \frac{\sigma_Z^2}{n} + O(n^{-2}), \quad \sigma_Z^2 := \text{Var}_q[Z],$$

both vanishing as $n \rightarrow \infty$: $\hat{L}_n$ is consistent, in fact $\sqrt{n}$ -consistent and asymptotically normal, $\sqrt{n}(\hat{L}_n - L_\infty) \rightarrow \mathcal{N}(0, L_\infty^2 \sigma_Z^2)$ in distribution, its bias shrinking at the same $O(1/n)$ rate as its variance. (Monte Carlo simulation with a well-specified $p_\theta = q$, 10^5 replicates per sample size $n \in \{5, \dots, 1000\}$, matches both formulas closely at every n tried, this note’s own verification rather than a claim requiring a citation.)

The second candidate is the raw product this note opened by rejecting, $L_{\text{raw}} := \prod_i p_\theta(Y_i | X_i) = \hat{L}_n^n = \exp(n\bar{Z}_n)$. Is it also an estimator of L_∞ ? Its expectation answers the question directly:

$$\mathbb{E}[L_{\text{raw}}] = (\mathbb{E}_q[e^Z])^n = \exp(n\Lambda(1)), \quad \Lambda(u) := \ln \mathbb{E}_q[e^{uZ}],$$

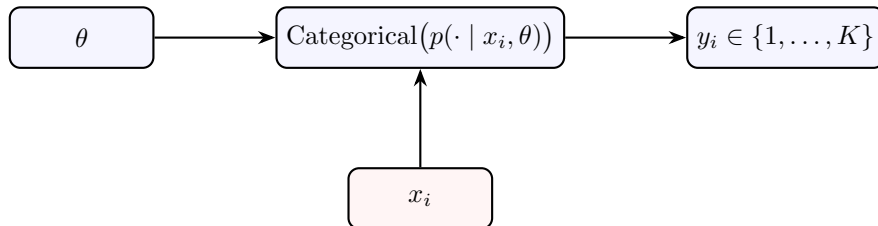
where Λ is exactly the cumulant generating function Cramér’s argument above is built from. For $\mathbb{E}[L_{\text{raw}}]$ to converge to L_∞ , or to anything fixed at all, $\Lambda(1)$ would need to equal 0 exactly, a knife-edge condition with no reason to hold for a generic p_θ . Absent it, $\mathbb{E}[L_{\text{raw}}]$ decays to 0 or explodes to ∞ exponentially fast in n . L_{raw} is not a biased or noisy estimator of L_∞ : it is not an estimator of it at all, the precise, quantitative version of Section 1’s opening observation that L_{raw} on 40 points and on 4,000 points are not numbers a single threshold could ever compare.

4 Two worked instantiations

The same construction, an exponentiated average log-density, lands on the reciprocal of a familiar error measure in both of the settings machine learning most often fits: a categorical classification model and a regression model under Gaussian noise.

4.1 Classification

A classifier’s generative model is the same graphical model as Section 1’s, specialized to a categorical outcome: x_i and θ set a probability vector over K classes, and y_i is drawn from it.



If Y is the true class and a model predicts $p(Y | X)$, $\exp(\mathbb{E}_{(X,Y)}[\ln p(Y | X)]) = \exp(-\text{CE})$ is exactly the geometric-mean probability the model assigns to the correct class, a reading no raw cross-entropy in nats offers directly. A cross-entropy of 0.693 nats, $\ln 2$, becomes $\exp(-0.693) \approx 0.5$: a model about as confident, on the correct class, as a coin flip. Equivalently, $\exp(\text{CE})$ is the *perplexity* the model assigns to the true class, the effective number of classes it is choosing among as far as its

confidence is concerned; a perplexity near 1 is a confident, well-calibrated model, a perplexity near K (the class count) is no better than guessing uniformly.

Take the $p = (0.5, 0.3, 0.15, 0.05)$ example Figure 1 already plotted, now read as one model’s predicted probabilities for one example, class 1 the true label. The model’s cross-entropy on that single example is $CE = -\ln p_1 = -\ln 0.5 \approx 0.693$ nats, exactly the coin-flip case above: the model assigns the true class only even odds, even though it is the single most likely class under its own prediction. Averaged over many examples rather than read off one, the same $\exp(-CE)$ becomes the geometric-mean probability the model assigns to whichever class turns out true each time. Figure 1’s right-hand panel is exactly that averaging machine turned into a dial: $\beta = 1$ is the ordinary, untilted cross-entropy read above, $\beta \rightarrow 0$ softens every prediction toward uniform confidence over all K classes regardless of the model’s own opinion, while $\beta \rightarrow \infty$ hardens every prediction toward its single most likely class, a temperature knob turning a probabilistic classifier into something closer to a bare arg max as it cools.

Figure 2 takes this from one hand-picked example to a genuine simulation: 45 points drawn from 3 overlapping classes, each assigned a probability for its own true class by a simple distance-to-centroid softmax model. Some points sit deep inside their own class’s territory and score high; others sit near a boundary between two classes and score low, the model genuinely unsure which label is right. L is the geometric mean of all 45 of those individual probabilities, not an arbitrary summary of them: the right-hand panel sorts them low to high and marks exactly where that geometric mean falls, visibly closer to the crowd of low scores than an arithmetic mean would sit, the same pull toward the worst cases Jensen’s inequality already predicted.

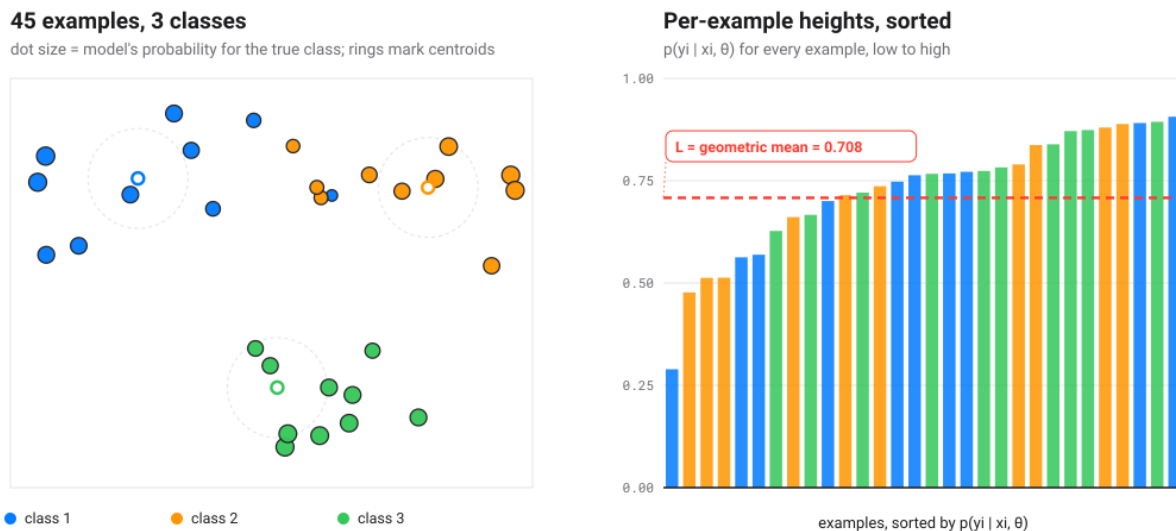


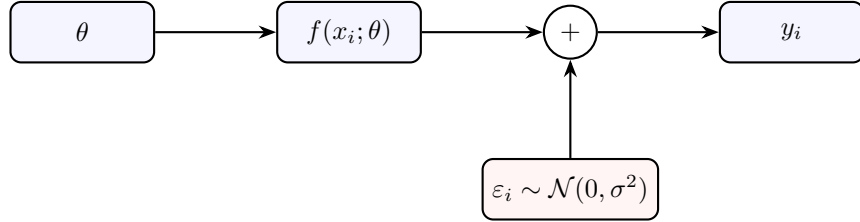
Figure 2: Left: 45 simulated examples from 3 overlapping classes (dashed rings mark the generating centroids), dot size showing the model’s probability for each example’s own true class, small near a class boundary, large deep inside a class’s territory. Right: those same 45 probabilities sorted from low to high, with their geometric mean, this note’s L , marked.

A single scalar, though, whether CE , $\exp(-CE)$, or a perplexity, only ever answers “how confident, on average”, never “how often actually right”. A model can be confidently wrong, assigning high probability to an incorrect class. CE penalizes exactly that overconfidence more harshly than

a small, honest uncertainty would be penalized, the same asymmetry a proper scoring rule is built to have: guessing 99% on a class that turns out false costs far more in nats than guessing 60% ever would. That asymmetry, not merely the average confidence level, is why cross-entropy rather than raw classification accuracy is the quantity fit by gradient descent in essentially every modern classifier.

4.2 Regression under Gaussian noise

Let $y_i = f(x_i; \theta) + \varepsilon_i$ with ε_i i.i.d. $\mathcal{N}(0, \sigma^2)$, f any regression function and θ its parameters, the generic conditional density of Section 1 specialized to one particular family:



$$p(y_i | x_i, \theta) = (2\pi\sigma^2)^{-1/2} \exp\left(-\frac{(y_i - f(x_i; \theta))^2}{2\sigma^2}\right) \quad (1)$$

so that, writing $\text{MSE}(\theta) := \frac{1}{n} \sum_i (y_i - f(x_i; \theta))^2$,

$$L(\theta, \sigma^2) = (2\pi\sigma^2)^{-1/2} \exp\left(-\frac{\text{MSE}(\theta)}{2\sigma^2}\right). \quad (2)$$

Figure 3 draws this density literally rather than only in symbols: a small bell curve at a few chosen x positions, each one the conditional density $p(\cdot | x_i, \theta)$ turned on its side, its width set by σ and its center riding along the fitted line. The marked dot on each bell is that curve's own height at the point actually observed, $p(y_i | x_i, \theta)$, tall where the fit lands close to the data, short where it does not. L is nothing more than the geometric mean of those heights taken over every observation, not only the four marked ones, the exact continuous counterpart of Figure 2's sorted bars: a curve fit closely almost everywhere posts mostly tall heights and a large L , a curve that wanders posts some short ones and pulls the geometric mean down, one bad miss doing more damage to a geometric mean than to an arithmetic one, the same asymmetry Section 2's Jensen bound already flagged.

At the maximum-likelihood variance $\hat{\sigma}^2 = \text{MSE}(\theta)$,

$$\exp(\mathbb{E}[\ln p]) = \frac{1}{\sqrt{2\pi e} \sqrt{\text{MSE}(\theta)}}, \quad (3)$$

the reciprocal of the root-mean-square error up to a universal constant $\sqrt{2\pi e}$, the exact continuous counterpart of Section 4.1's reciprocal-perplexity reading: in both cases $\exp(\mathbb{E}[\ln p])$ is the reciprocal of an error measure native to the prediction task, perplexity for a categorical one, root-mean-square error for a continuous one. This is no coincidence. Writing $\ell := \ln L$ and, following the convention information criteria use, $\text{PPL} := \exp(-2\ell)$, the identity $\exp(\mathbb{E}[\ln p]) = 1/\sqrt{\text{PPL}}$ holds in general, Gaussian or not, a direct consequence of the two definitions rather than a property specific to either worked case. `elbow-helper`'s own `ELBOW-en.tex` takes this instantiation further: f there ranges over single lines, broken lines and k -segment piecewise lines. PPL is turned into a genuinely bounded, worst-case-anchored score rather than left as a bare reciprocal-RMSE reading.

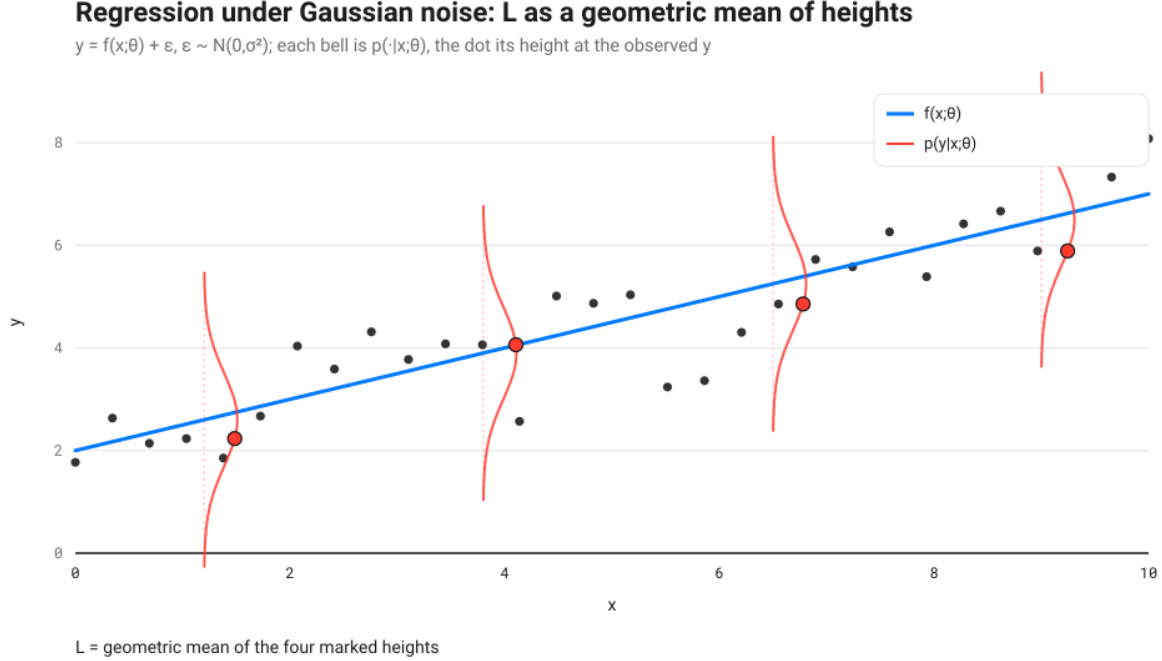


Figure 3: Synthetic regression data, $y = 2 + 0.5x + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, 0.9^2)$, with the fitted line and four conditional-density curves $p(\cdot | x, \theta)$ drawn at selected x positions. Each dot marks that curve’s own height at the value actually observed there. L is the geometric mean of every such height, not only the four marked: high where the fit lands close to the data, low where a point falls far from the curve’s center.

5 A practical conclusion: bounding the likelihood

Every quantity this note built, CE, ℓ , PPL, is a rescaling of the same underlying number. Every one of those rescalings shares the same first problem: taken raw, it has no ceiling, so no single threshold on it means the same thing across two different fits. Fixing that takes anchoring against a real, achievable worst case rather than an arbitrary or unreachable one, a move classification needs once and regression needs twice over, for a reason worth making precise rather than waved at.

Classification’s $\text{CE}(\theta) \geq 0$ already has a clean floor: zero exactly at a perfectly confident, correct prediction, nothing special has to be done to reach it. Section 2’s temperature sweep already named the natural ceiling to anchor against. At $\beta \rightarrow 0$, hot and maximally agitated, any model’s confidence flattens to uniform over the K classes regardless of what it actually believed. A model that predicts uniformly earns exactly $\text{CE} = \ln K$ on any true class whatsoever, the one cross-entropy value that does not depend on which class turns out right. That is not an arbitrary reference: it is what refusing to use any of the model’s own information costs, in nats, no matter the label, so

$$Q_{\text{classif}}(\theta) := 1 - \frac{\text{CE}(\theta)}{\ln K}$$

reads 1 for a perfectly confident, correct model, 0 for one no better than uniform guessing, and negative for one actively misleading, confidently wrong more often than confidently right, a signal this note sees no reason to hide by clipping it away.

Regression’s own log-likelihood has no such floor. That is the second problem, distinct from the ceiling both cases share. A continuous density can exceed 1, so profiling σ^2 at its best-supported value $\hat{\sigma}^2 = \text{MSE}(\theta)$ sends $\ell(\theta, \hat{\sigma}^2)$ to $+\infty$, not to a clean 0, exactly at a perfect fit, the classical likelihood degeneracy behind BIC’s whole reason to exist. Exponentiating repairs that first: $\text{PPL}(\theta) := \exp(-2\ell(\theta, \hat{\sigma}^2)) = 2\pi e \cdot \text{MSE}(\theta) \geq 0$, zero exactly at a perfect fit, still unbounded above but now starting from a floor worth building on. `elbow-helper`’s own `ELBOW-en.tex` supplies the second move, the ceiling: not a formula but a genuine worst case drawn from the data itself, the single *observed* point y_{bad} hardest to predict everything else from, giving $\text{MSE}_{\text{worst}}$ the exact role $\ln K$ plays above,

$$Q_{\text{reg}}(\theta) := 1 - \frac{\text{MSE}(\theta)}{\text{MSE}_{\text{worst}}} = 1 - \frac{\text{PPL}(\theta)}{\text{PPL}_{\text{worst}}},$$

the two ratios equal because PPL and MSE carry the same constant $2\pi e$ on both the numerator and the denominator.

Two different worst cases answer the same question in the only two ways this note’s own generative model allows. A K -category classifier has a fixed, label-independent alternative built into its own structure, uniform guessing, and never has to look at the data to find one. A continuous regressor has no such built-in alternative: nothing about the model family alone says what a bad prediction looks like, so the worst case has to be borrowed from the very sample the model is trying to fit. Once that asymmetry is priced in rather than papered over, CE and MSE, numbers that only ever meant something next to another run of the same model on the same data, become Q_{classif} and Q_{reg} , numbers that mean the same thing on any dataset, of any size, in either language this note is written in.

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